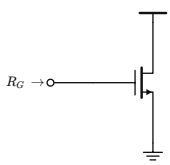
ECE 342 · Exam 2 stage sheet $i_d = g_m v_{gs} + v_{ds}/r_{ds}$ (D→S), NMOS and PMOS alike · $g_m = \frac{2ID}{V_{ov}} = \mu C_{ox}\frac{W}{L}V_{ov} = \sqrt{2\mu C_{ox}\frac{W}{L}I_D}$

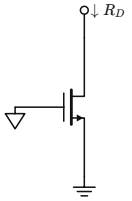
Small-signal circuit

- V_{DD}, bias V_B, ▽ → **incremental ground**. DC current source → **open**. ∞ capacitor → short.
- Gate–source shorted by a **wire**: $V_{GS} = 0$, device **off** → open. By a big **cap**: biased, $v_{gs} = 0$ → r_{ds}.
- $1\,\text{mA}$ at $V_{ov} = 0.2$: $g_m = 10\,\text{mS}$, $1/g_m = 100\,\Omega$. Same I: $V_{ov} \propto \frac{1}{\sqrt{W/L}}$, $g_m \propto \sqrt{W/L}$.

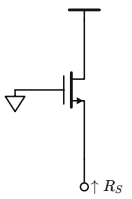
Looking in (others at incremental ground)



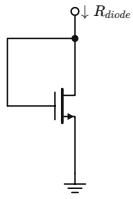
Gate: $R_G = \infty$
Never loads the node it hangs on.



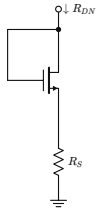
Drain: $R_D = r_{ds}$
$v_{gs} = 0$: a current source, imperfect by r_{ds}.



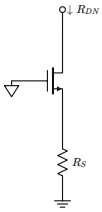
Source: $\frac{1}{g_m} \parallel r_{ds} \approx \frac{1}{g_m}$
Push it up v: $v_{gs} = -v$, $g_m v$ flows in. Sources are low.



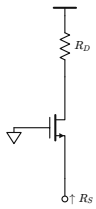
Diode (G = D): $\frac{1}{g_m} \parallel r_{ds} \approx \frac{1}{g_m}$
PMOS diode: the same. Stacked diodes add.



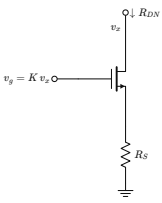
Diode on R_S: $R_S + \frac{1}{g_m} \parallel r_{ds}$
Gate follows drain ($K = 1$): no magnification. Not $\approx R_S$!



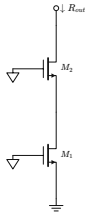
Degenerated drain: $r_{ds} + R_S + g_m r_{ds} R_S \approx r_{ds}(1 + g_m R_S)$
R_S looks $g_m r_{ds}$ times bigger from the drain.



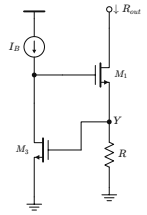
Source, R_D on drain: $\frac{r_{ds} + R_D}{1 + g_m r_{ds}} \approx \frac{1}{g_m} + \frac{R_D}{g_m r_{ds}}$
R_D looks $g_m r_{ds}$ times smaller. **Ideal current source on the drain: ∞.**



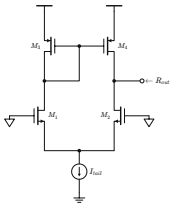
K rule (gate moves Kv_x):
$\frac{r_{ds} + R_S + g_m r_{ds} R_S}{1 + g_m r_{ds} K} \approx \frac{R_S + 1/g_m}{K}$
$K = 0$: degenerated drain; $K = 1$: diode. $R_S = 0$: $r_{ds} \parallel \frac{1}{K g_m}$. Divider: $K = \frac{R_b}{R_a + R_b}$.



Cascode: $r_{ds1} + r_{ds2} + g_{m2} r_{ds2} r_{ds1} \approx g_{m2} r_{ds2} r_{ds1}$
Degenerated drain with $R_S = r_{ds1}$. Work any stack bottom-up: $R_{next} = r_{ds} + R_x(1 + g_m r_{ds})$.



Regulated cascode: $r_{ds1} + R + g_{m1} r_{ds1}(1 + A)R$, $A = g_{m3} r_{ds3}$
The booster drives the gate the other way: v_{gs1} swings $(1 + A)\times$.



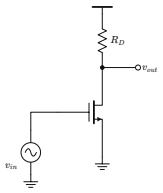
Diff pair, mirror load: $R_{out} \approx r_{ds2} \parallel r_{ds4}$
Test current: down M_2, up M_1 (it sees $\approx 2r_{ds}$), onto the diode, copied by M_4: counts twice. Current-source loads instead: $r_{ds4} \parallel \left[r_{ds2} + (1 + g_{m2} r_{ds2})\frac{r_{ds1} + r_{ds3}}{1 + g_{m1} r_{ds1}}\right] \approx r_{ds4} \parallel 3r_{ds}$.

Method for any node

- $R = R_{UP} \parallel R_{DN} \parallel$ (transistors whose gate the node drives). The smallest branch wins: spot the $1/g_m$ ones first.
- Each branch from its far end toward the node, replacing finished pieces by one resistance.
- A gate the test voltage reaches: find K (divider, a diode in a stack $K = \frac{R_{below}}{R_{branch}}$). Feedback through a mirror: $R_{branch}/(1 + \text{extra current per unit } i)$.
- Keep an r_{ds} when nothing smaller is in parallel with it; drop it next to $1/g_m$.
- **Box** R_{UP}, R_{DN} separately: partial credit.

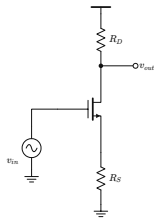
Single stages

$A_v = -G_M R_{out}$. G_M: output shorted, current pulled **down** into the stage per v_{in} (add every path). R_{out}: v_{in} dead, $R_{UP} \parallel R_{DN}$.



CS: $G_M = g_m$, $R_{out} = R_D \parallel r_{ds}$, $A_v = -g_m(R_D \parallel r_{ds})$

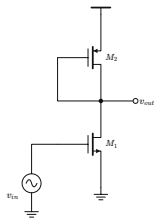
$v_{in} \uparrow \rightarrow$ pulls more $\rightarrow v_{out} \downarrow$.



CS + R_S: $G_M = \dfrac{g_m}{1 + (g_m + g_{ds})R_S}$, $R_{out} = R_D \parallel r_{ds}(1 + g_m R_S)$

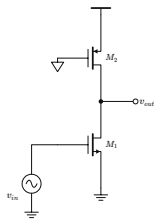
$A_v \approx -\dfrac{g_m R_D}{1 + g_m R_S} \rightarrow -\dfrac{R_D}{R_S}$

Source on an ideal current source: contributes $G_M = 0$. Diode in the source: $\frac{g_{m1}g_{m3}}{g_{m1}+g_{m3}}$.



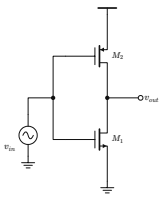
Diode load: $A_v \approx -\dfrac{g_{m1}}{g_{m2}} = -\sqrt{\dfrac{\mu_1 (W/L)_1}{\mu_2 (W/L)_2}}$

Same current. NMOS over a PMOS diode: $\mu_n/\mu_p = 2$ under the root.



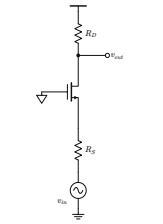
Current-source load: $A_v = -g_{m1}(r_{ds1} \parallel r_{ds2}) \approx -\frac{1}{2} g_m r_{ds}$

The most one transistor gives.



CMOS inverter: $A_v = -(g_{m1} + g_{m2})(r_{ds1} \parallel r_{ds2})$

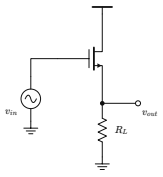
Same signal on both gates: the currents add.



CG: $A_v = +\dfrac{R_D \parallel R_{dev}}{R_S + \frac{1}{g_m} \parallel r_{ds}} \approx +\dfrac{R_D}{R_S + 1/g_m}$

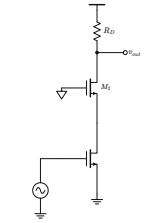
$R_{in} = \dfrac{r_{ds} + R_D}{1 + g_m r_{ds}}$, $R_{out} = R_D \parallel \underbrace{(r_{ds} + R_S + g_m r_{ds} R_S)}_{R_{dev}}$

Non-inverting ($G_M < 0$). $R_S = 0$: $(1 + g_m r_{ds})\frac{R_D}{R_D + r_{ds}}$.



CD (follower): $A_v = \dfrac{R_L \parallel r_{ds}}{(R_L \parallel r_{ds}) + 1/g_m}$, $R_{out} = R_L \parallel \frac{1}{g_m} \parallel r_{ds}$

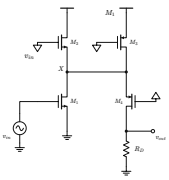
A divider with $1/g_m$. Ideal current source: $\frac{g_m r_{ds}}{1 + g_m r_{ds}} \approx 1$, $V_{OUT} = V_{IN} - V_{GS}$.



Cascode: $G_M = \dfrac{g_{m1} r_{ds1}(1 + g_{m2} r_{ds2})}{r_{ds1} + r_{ds2} + g_{m2} r_{ds1} r_{ds2}} \approx g_{m1}$

$R_{out} = R_D \parallel (r_{ds1} + r_{ds2} + g_{m2} r_{ds2} r_{ds1})$, $A_v \approx -g_{m1} R_D$

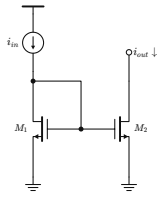
Multi-transistor



Folded cascode: $R_X = r_{ds1} \parallel r_{ds3} \parallel \frac{1}{g_{m2}}$, $G_M = g_{m1} \dfrac{R_X}{R_X + 1/g_{m4}}$

$R_{out} = R_D \parallel [r_{ds4} + R_X(1 + g_{m4} r_{ds4})]$

Current division at X; then a CG degenerated by R_X.



Mirror: $i_{out} = \dfrac{g_{m2}}{g_{m1}} i_{in} = \dfrac{(W/L)_2}{(W/L)_1} i_{in}$, diode node $= i_{in}/g_{m1}$

The copy flows the same way. CS → PMOS diode → mirror into R_L: $A_v = +\frac{g_{m1}g_{m3}}{g_{m2}} R_L$.

- **Chains:** $A_v = A_1 A_2$; find v_1 with the next stage attached (gate: free; source/diode: $1/g_m$ in parallel).
- **Two paths:** short the output, add every transistor's current: $G_M = G_{M,a} + g_{m,b} A_1$. A PMOS gate fed an inverted copy fights an NMOS fed v_{in}.
- **Current division:** into a $1/g_m$ branch from R_X: share $\frac{R_X}{R_X + 1/g_m}$.
- **Signs:** CS-type stages invert, CG and CD do not. Check: "$v_{in} \uparrow \rightarrow$ current … $\rightarrow v_{out}$ …".

DC (Problem 1)

- Mirror (same V_{GS}): $I \propto W/L$. KCL: resistors take the difference of mirrored currents.
- Saturation: NMOS $V_D \geq V_G - V_T$; PMOS $V_D \leq V_G + |V_T|$; edge = equality. Diodes are always saturated.
- $K = \frac{1}{2}\mu C_{ox}\frac{W}{L}$; $\mu_p C_{ox} = 50$ is half of $\mu_n C_{ox} = 100\,\mu\text{A/V}^2$: a PMOS needs $2\times$ the W/L.
- Numeric gains: DC only as far as g_m; ∞ caps short; watch an input divider into a $1/g_m$ node.